

Case Study on the Analysis of SDLC Models Applied to a Real-World Software Application (Bus Reservation System)

Application: Bus Reservation System

Aim

To analyze the SDLC models used in the development of a Bus Reservation System and identify the most suitable SDLC model.

Objectives

- Study different SDLC models.
- Analyze the development process of a Bus Reservation System.
- Compare advantages and limitations of SDLC models.
- Recommend the most suitable SDLC model.

Overview

A Bus Reservation System allows passengers to search buses, select seats, book tickets, make online payments, cancel bookings, and download e-tickets. It also enables operators and administrators to manage buses, routes, schedules, fares, and reports.

Functional Requirements

Passenger Module

- Registration
- Login
- Search Buses
- Seat Selection
- Book Tickets
- Online Payment
- Cancel Ticket
- Booking History

Bus Operator Module

- Manage Buses
- Manage Routes
- Update Schedules
- Manage Seat Availability
- View Bookings

Admin Module

- User Management
- Operator Management
- Reports
- Analytics

SDLC Models Analysis

Model	Suitability	Advantages	Limitations
Waterfall	Low	Simple	Hard to change requirements
Prototype	Medium	Better requirement understanding	Higher cost
Spiral	High	Risk management	Complex
Agile	Very High	Fast iterations & feedback	Needs experienced team
DevOps	Very High	CI/CD	Infrastructure complexity

Why Agile is Best

- Frequent requirement changes
- Quick feature releases
- Continuous customer feedback
- Regular bug fixes
- Mobile app updates

SDLC Phases

Requirement Analysis

SRS, passenger, operator, payment, routes

System Design

UI, database, APIs, architecture

Development

Login, search, booking, payment, admin modules

Testing

Unit, Integration, System, UAT

Deployment

Cloud, Web, Mobile

Maintenance

Bug fixes, security updates, enhancements

Conclusion

Agile with DevOps is the most suitable SDLC approach for a Bus Reservation System because it supports continuous development, testing, deployment, flexibility, and customer satisfaction.

References

- Pressman, R. S., & Maxim, B. R. (2020). *Software Engineering: A Practitioner's Approach* (9th Edition). McGraw-Hill.
- RedBus
<https://www.redbus.in/>
- AbhiBus
<https://www.abhibus.com/>
- Oracle Database Documentation
<https://docs.oracle.com/en/database/>
- <https://www.ibm.com/topics/software-development-life-cycle>
- <https://docs.aws.amazon.com>